

## Dr. Jyoti Panda

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| CONTACT INFORMATION  | Mechanical Engineering Department<br>University of North Carolina, Charlotte<br>North Carolina, USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Phone: +1-(551) 260-1943<br>Email: <a href="mailto:jppanda.iit@gmail.com">jppanda.iit@gmail.com</a><br>Homepage: <a href="https://dr-jppanda.github.io">https://dr-jppanda.github.io</a> |
| PROFESSIONAL SUMMARY | Accomplished computational thermo-machanical modeling researcher with strong analytical, organizational, and project management skills.<br><br>Demonstrated expertise in developing high-fidelity computational, experimental and data-driven models for heat exchanger, microelectronic, aerospace and mechanical systems involving fluid flow and heat transfer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                          |
| RESEARCH INTERESTS   | Computational modeling and simulation, Fluid mechanics, Heat Transfer, Computational Fluid Dynamics, Thermal management, Experiments, Microelectronic heat transfer, Heat exchangers, Thermo-electrics, Joule Heating, Fluid mixing, Acoustics, Microfluidics, Rarefied gas flows, Turbulent Flows, Data-driven Modeling, Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                          |
| EXPERIENCE           | <b>Complex Fluids-Multiphase Lab, UNC Charlotte, USA</b> May 2025 to Present<br>Post-Doctoral Research Fellow<br>Funding Agency: NASA, NSF <ul style="list-style-type: none"><li>• Conducted heat exchanger experiments to collect acoustic and thermofluidic data under varying operating conditions.</li><li>• Developed machine learning models to predict heat transfer characteristics from acoustic signals.</li><li>• Validated predictions against experiments, improving understanding of condensation dynamics.</li></ul> <b>University of Notre Dame, USA</b> Oct 2023 to May 2025<br>Post-Doctoral Research Associate<br>Funding Agency: Defense Advanced Research Projects Agency (DARPA), USA <ul style="list-style-type: none"><li>• Conducted TCAD simulations to model Joule heating and performed thermo-electric modeling of transistors in microelectronic circuits</li><li>• Predicted the thermal field in large FinFET transistor arrays using both JAX-based BTE and Fourier simulations.</li></ul> <b>Research Center for Aircraft Core Technology, S. Korea</b> Aug 2022 to Sep 2023<br>Senior Researcher<br>Funding Agency: NRF-South Korea, AOARD-AFSOR-USA <ul style="list-style-type: none"><li>• Performed DSMC simulations of rarefied gas flows using SPARTA and Bird's code.</li><li>• Conducted DNS simulations of decaying compressible turbulence and shock-turbulence interactions.</li><li>• Investigated shock-boundary layer interactions on high-performance computing servers running Linux</li></ul> <b>Indian Institute of Technology Kharagpur, India</b> Mar 2019 to June 2021<br>Research Associate<br>Funding Agency: NRB, Defense Research and Development Organisation, India <ul style="list-style-type: none"><li>• Conducted CFD simulations in Ansys Fluent to investigate drag reduction strategies for underwater vehicles using micro-dimples and grooves.</li><li>• Analyzed flow behavior to quantify the impact of surface modifications on drag and hydrodynamic performance.</li><li>• Implemented machine learning models to enable fast and accurate prediction of the flow field, reducing computational time.</li></ul> |                                                                                                                                                                                          |

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| EDUCATION                             | <b>Indian Institute of Technology Kharagpur</b> , Kharagpur, WB, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                        |
|                                       | Ph.D., Computational Fluid Dynamics (Turbulence Modeling), Aug 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                        |
|                                       | <ul style="list-style-type: none"> <li>• Thesis Topic: <i>Pressure Strain Correlation Modeling for Turbulent Flows</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |
|                                       | <b>Indian Institute of Engineering Science and Technology, Shibpur</b> , WB, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                        |
|                                       | M.E., Mechanics of Fluids (CFD-Microfluidics), June 2015 <ul style="list-style-type: none"> <li>• Department: Aerospace Engineering and Applied Mechanics</li> <li>• Thesis Topic: <i>Electroosmotic Mixing and Joule Heating in Microchannels</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                      |                        |
|                                       | <b>National Institute of Electronics and Information Technology, India</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                        |
|                                       | P.G. (Post Graduate), Data Engineering, Online Course, Nov 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                        |
|                                       | <ul style="list-style-type: none"> <li>• Configuring Linux Platform for Data Engineering</li> <li>• Data Analytics and Machine Learning using Tensorflow and Keras</li> <li>• Big Data Analytics with Hadoop and Apache Spark</li> </ul>                                                                                                                                                                                                                                                                                                                                                                           |                        |
|                                       | <b>Biju Patnaik University of Technology, Rourkela</b> , ODISHA, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                        |
|                                       | B.Tech., Mechanical Engineering, June 2012 <ul style="list-style-type: none"> <li>• Project: <i>Design and fabrication of a pneumatic material handling system</i></li> <li>• CGPA 8/10</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |                        |
| PROGRAMMING<br>AND SOFTWARE<br>SKILLS | Programming:<br>Python, C/C++, Fortran                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                        |
|                                       | Software/Codes:<br>Docker and Singularity container<br>ANACONDA (For python environments)<br>JAX-Fluids (DNS and LES) (Parallel CUDA, Python)<br>STREAMS (DNS) (Parallel MPI/CUDA, FORTRAN)<br>GiftBTE (Boltzmann transport equation) (submicron thermal transport)<br>OpenFOAM (RANS, LES, DSMC) (Parallel MPI, C++)<br>SU2 (Parallel MPI, C++)<br>ANSYS Fluent (RANS and RSTM)<br>Gmsh/Ansys-workbench/Gambit (Meshing)<br>SPARTA and Prof. Bird's code (DSMC)<br>TensorFlow and Keras (Deep learning)<br>TensorFlowFoam(Linking neural network models with OpenFoam)<br>Scikit Learn (Machine Learning), MATLAB |                        |
| EXPERIMENTAL<br>WORK                  | Instrument: Acoustic Doppler Velocimeter<br>Principle of operation: Doppler Shift<br>Measured parameters: Three fluctuating turbulent velocity components in grid generated turbulence with and without mean strain<br>Location: Recirculating water tank, Ship Hydrodynamics Lab, IIT Kharagpur                                                                                                                                                                                                                                                                                                                   |                        |
| SCHOLARSHIPS                          | <ul style="list-style-type: none"> <li>• MHRD government of India fellowship for doctoral studies , India</li> <li>• MHRD government of India fellowship for PG studies , India</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                         | 2015-2018<br>2013-2015 |

1. **Panda, J.**, Shang, W., Liu, Y., Zhou, J., Liang, J., Wang, J.X. and Luo, T.,“Scalable non-diffusive thermal analysis of field-effect transistors using neural operators” 2026, *APL Electronic Devices*, 2(1).
2. Yan, L., Li, J.,**Panda, J.**, Kharangate, C.R. and Sun, Y.,“Prediction of local heat transfer characteristics in flow condensation via acoustic Emission sensing” 2026, *Applied Thermal Engineering*, p.129120. (Impact Factor: 6.9) (Q1- Energy Engineering and Power Technology)
3. **Panda, J.**, Warrior, H.,“Numerical studies on drag reduction of an axisymmetric body of revolution with antiturbulence surface” 2021, *ASME Journal of Offshore Mechanics and Arctic Engineering*, 143(6), p.064501. (Impact Factor: 1.355) (Q2 Energy)
4. Jhou, J, Shang, W, Li, Ruiyang, Liu,Yi ,**Panda, J.**,...,Wang, J , Luo, T,“Physics-Informed Neural Networks with Hard-Encoded Angle-Dependent Boundary Conditions for Phonon Boltzmann Transport Equation”2025, *Materials Today Physics*, 59, pp.101922.(Q1 Energy)
5. Shang, W, Jhou, J,**Panda, J.**,Wang, J , Luo, T,“JAX-BTE: a GPU-accelerated differentiable solver for phonon Boltzmann transport equations”2025, *npj Computational Materials*, 11(1), pp.1-12.(Q1 Computer Science Applications)
6. **Panda, J.**, Warrior, H.,“Modeling pressure strain correlation for turbulent flows using deep neural networks” 2021, *Proceedings of the Institution of mechanical engineers, Part C: Journal of Mechanical Engg. Science*. (Impact Factor: 1.762). (Q2 Mechanical Engineering)
7. **Panda, J.**, Warrior, H.,“Data-driven prediction of complex flow field over an axisymmetric body of revolution using Machine Learning” 2022, *ASME Journal of Offshore Mechanics and Arctic Engineering*. (Impact Factor: 1.355) (Q2 Energy)
8. Mitra, A. **Panda, J.**, Warrior, H.,“The effects of free stream turbulence on the hydrodynamic characteristics of an AUV hull form” 2019. *Ocean Engineering, Vol. 174 (2) / 148-158*. (Impact Factor: 3.795) (Q1 Environmental engineering)
9. Mitra, A., **Panda, J.**, Warrior, H.,“Experimental and numerical investigation of the hydrodynamic characteristics of Autonomous Underwater Vehicles over seabeds with complex topography”2020. *Ocean Engineering, Volume 198, 106978* . (Impact Factor: 3.795) (Q1 Environmental engineering)
10. **Panda, J.**, Warrior, H.,“A representation theory based model for the rapid pressure strain correlation of turbulence”2018, *ASME Journal of Fluids Engg.*, Vol. 140 / 081101-1. (Impact Factor: 1.995) (Q2 Mechanical Engineering)
11. **Panda, J.**, Warrior, H.,“Evaluation of machine learning algorithms for predictive Reynolds stress transport modeling” 2021, *Acta Mechanica Sinica* , (Impact Factor: 1.975) (Q2 Computational Mechanics)
12. **Panda, J.**, Kumar, B., Kumar, A.,Patil, A.,“Influence of twisted tape length on the thermal performance of a heat exchanger tube” 2022, *Numerical Heat Transfer, Part A: Applications*, (Impact Factor: 2.928) (Q2 Condensed matter physics)
13. **Panda, J.**, Kumar, B., Patil A., Kumar M. “Machine learning assisted modelling of thermohydraulic correlations for heat exchangers with twisted tape inserts, 2023, *Acta Mechanica Sinica* (Accepted), (Impact Factor: 1.975) (Q2 Computational Mechanics)

14. **Panda, J.**, Warrior, H., Maity, S., Mitra, A., Sasmal, K., “An improved model including length scale anisotropy for the pressure strain correlation of turbulence” 2017, *ASME Journal of Fluids Engineering*, Vol. 139 / 044503-1. (Impact Factor: 1.995) (Q2 Mechanical Engineering)
15. **Panda, J.**, “A review of pressure strain correlation modeling for Reynolds stress models ” 2019, *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*. DOI: <https://doi.org/10.1177/095440-6219893397>. (Impact Factor: 1.762) (Q2 Mechanical Engineering)
16. **Panda, J.**, Mitra, A., Warrior, H., “A review on the hydrodynamic characteristics of autonomous underwater vehicles” 2020. *Proceedings of the Institution of Mechanical Engineers, Part M: Journal of Engineering for the Maritime Environment*, DOI: <https://doi.org/10.1177/1475090220936896>. (Impact Factor: 1.389) (Q2 Mechanical Engineering)
17. **Panda, J.**, “A reliable pressure strain correlation model for complex turbulent flows” 2020. *Journal of applied fluid mechanics*, Vol. 13, No. 4, pp. 1167-1178. (Impact Factor: 1.783) (Q3 Condensed matter physics)
18. **Panda, J.**, Sasmal, K., Maity, S., Warrior, H., “A Simple Nonlinear Eddy Viscosity Model for Geophysical Turbulent Flows” 2020, *Journal of Applied Fluid Mechanics*, 14(3). (Impact Factor: 1.783) (Q3 Condensed matter physics)
19. Das A., Das, SR., **Panda, J.**, Dey A., Gajrani KK., Somani N., Gupta N. “Machine learning based modelling and optimization in hard turning of AISI D6 steel with advanced AlTiSiN coated carbide inserts, 2022, *Surface Review and Letters*. (Impact Factor: 1.303) (Q3 Materials chemistry)
20. Somani, N., Walia, A., Gupta, N., **Panda, J.**, Das, A., Das S. “Data driven surrogate model-based optimization of the process parameters in electric discharge machining of D2 steel using Cu-SiC composite tool” 2023, *Revista de Metalurgia*. (Impact Factor: 0.8) (Q3 Metals and Alloys)
21. **Panda, J.**, Mitra, A., Joshi, A., Warrior, H., “Experimental and numerical analysis of grid generated turbulence with and without mean strain” 2018, *Experimental Thermal and Fluid Science*, Vol. 98 (11) / 594-603. (Impact Factor: 3.232) (Q1 Aerospace engineering)
22. **Panda, J.**, Handique J., Warrior, H., “Mechanics of drag reduction of an axisymmetric body of revolution with shallow dimples” 2022, *Proceedings of the iMech, Part M: Journal of Engineering for Maritime Environment*. (Impact Factor: 1.389) (Q2 Mechanical Engineering)
23. **Panda, J.** “Machine learning for Naval Architecture Ocean and Marine Engineering” 2023, *Journal of Marine Science and Technology*. (Impact Factor: 2.005) (Q1 Mechanical Engineering)
24. Mitra, A., **Panda, J.**, Warrior, H., “The hydrodynamic characteristics of autonomous underwater vehicles in rotating flow fields” 2023, *Proceedings of the iMech, Part M: Journal of Engineering for Maritime Environment* (Q2 Mechanical Engineering).

SUBMITTED  
JOURNAL  
PUBLICATIONS

1. **Panda, J.**, Yan, L, Zeng, Y and Hu, H and Sun, Y, “Conditional Generative Modeling of Condenser Dynamics Using Acoustic Signatures, *AI Thermal Fluids* (2025).

1. **Panda, J.**,...,Wang, J , Luo, T, “Scalable Thermal Analysis of Field-Effect Transistors Using Neural Operators” 2025. *18th U.S. National congress on computational mechanics, Chicago, Illinois, July 20-24, 2025.*
2. **Panda, J.**,...,Wang, J , Luo, T, “Data-Driven Prediction of Thermal Field in Field-Effect Transistors Using Deep Neural Networks” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024).*
3. Zhou J, **Panda, J.**,...,Wang, J , Luo, T, “ Jax-Bte: A Differentiable Hybrid Neural Solver for Deep Learning Accelerated Thermal Modeling of Nanoelectronics ” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024).*
4. Shang W, **Panda, J.**,...,Wang, J , Luo, T, “Physics-Integrated Hybrid Machine Learning Model for Phonon Bte ” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024) .*
5. Zhou J, **Panda, J.**,...,Wang, J , Luo, T, “ Physics-Informed Neural Networks for Transistor Thermal Modeling Using Phonon Boltzmann Transport Equation ” 2024. *ASME 2024 Heat Transfer Summer Conference (SHTC2024).*
6. J Zhou, R Li, W Shang, B Zhang, Z Xu, **Panda, J.**, J Wang, T Luo, T Beechem, E Walker, “Efficient Physics-Informed Deep Learning Model for Multiscale Thermal Analysis” 2024. *Government Microcircuit Applications and Critical Technology (GOMACTech) Conference.*
7. **Panda, J.**, Myong, R. S., “Subgrid Modeling for Large Eddy Simulation of Shock Boundary Layer Interaction Using Machine Learning” 2023. *The 14th Asian Computational Fluid Dynamics Conference, CSIR, National Aerospace Laboratories, Bengaluru, India.*
8. **Panda, J.**, Sengupta, B., Myong, R. S., “Direct numerical simulation of shock turbulence interaction with bulk viscosity effects” 2023. *The 34th International Symposium on Shock Waves, Daegu, Korea.*
9. **Panda, J.**, Gupta, S., Pal, D., “Computational Analysis of Liquid-Liquid Mixing In a T-Shaped Serially Connected Converging-Diverging microchannel” 2014. *59th Congress of ISTAM, IIT Kharagpur, India.*
10. Mohapatra, P., **Panda, J.**, Pal, D., “Electro-osmotic Flow and Mixing in a Micro-channel: A Numerical Study” 2014. *59th Congress of ISTAM, IIT Kharagpur, India.*
11. **Panda, J.**, Warrior, H., Maity, S., “Pressure Strain Correlation for decaying homogeneous turbulence” 2016. *Fluid Mechanics and Fluid Power Conference held at MNNIT Allahabad, India.*
12. Joshi, A., Warrior, H., **Panda, J.** “An Improved Single Point Closure Model Based on Dissipation Anisotropy for Geophysical Turbulent Flows” 2018. *Int. Conference on Oceanography held at Miami, USA.*
13. Gupta, S., **Panda, J.**, Nandi, N. “A Model Study of Free Vortex Flow” 2014. *ICTACEM Conference held at IIT Kharagpur, India.*

REFERENCES

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PERSONAL  
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